Authenticated Encryption: Why and How?

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Communication

Communication between geographically distant parties.

Content	Form	Applications
voice	phone	military
picture	FAX	business
computer file	email	politics
text	SMS	diplomacy
...	...	...

Message: a sequence of bits (bitstream).



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Two Aspects of Communication

- **Error-free communication.**
  - Reliable communication over a noisy channel.
  - Coding theory.
- **Secure communication.**
  - Private communication over an open (public) channel.
  - Cryptography.

We will be interested in cryptography.



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Sensitive Messages

Short messages.

- Headquarter to field unit: **Attack at dawn.**
- MD to broker: **Buy XYZ shares.**

Long messages.

- E-mails, phone coversations, text files, images (maps).

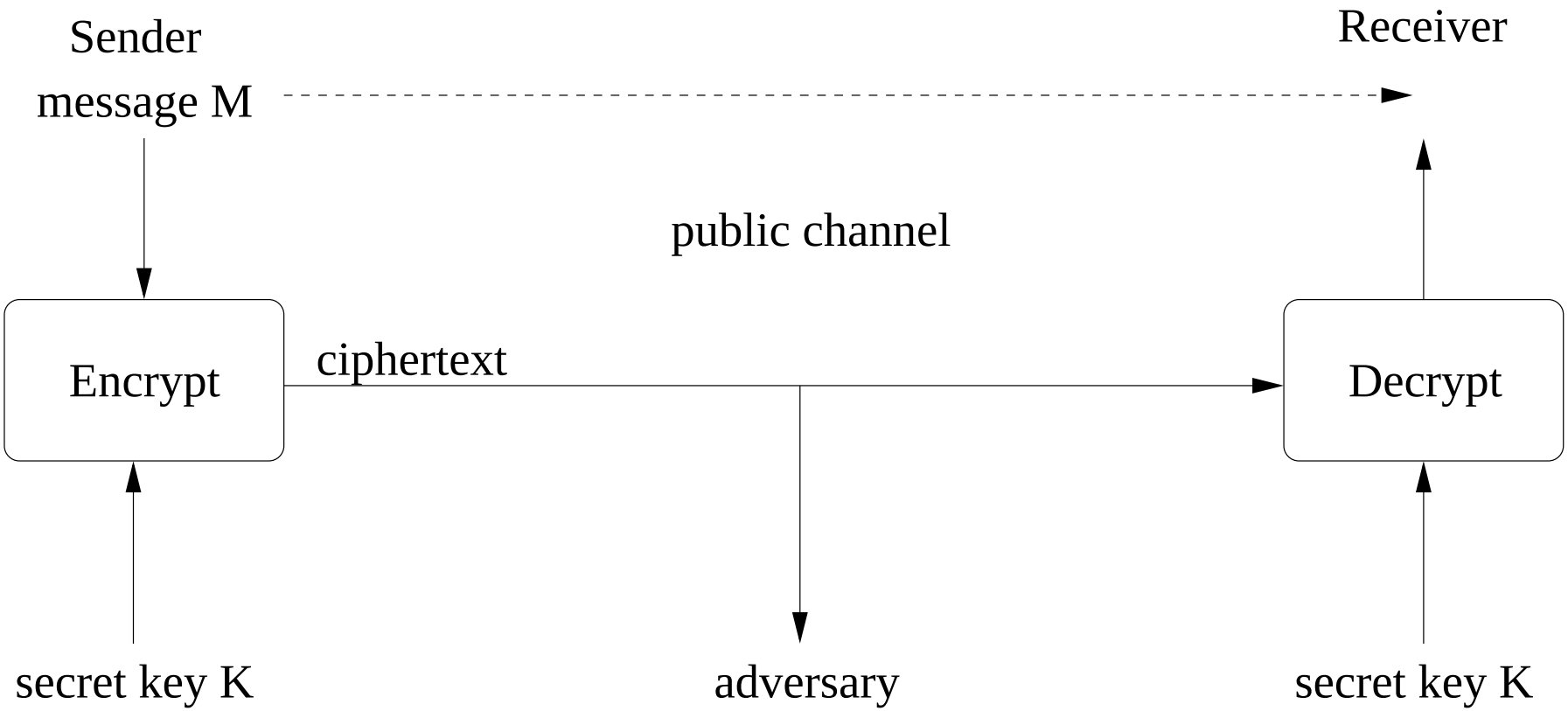


Communication Over the Internet

- No dedicated path from sender to receiver.
- Messages are broken into packets.
- Each packet consists of the following.
  - Body: actual data.
  - Header: identifying information for a packet.



Model of Encryption



Security Requirement: Confidentiality

Assumption: adversary is passive, i.e., does not modify the information on the channel.

- Ciphertext should not leak information about the message or the key.
- Adversary may have access to previous message-ciphertext pairs.
  - **Ciphertext only attack:**
    the attacker has access to only ciphertext(s);
    goal: find (one of) the message(s) and/or the key.
  - **Known plaintext attack:**
    the attacker *knows* $(P_1, C_1), \ldots, (P_t, C_t)$;
    goal: find the key or find P^* corresponding to a "new" C^*.
  - **Chosen plaintext attack:**
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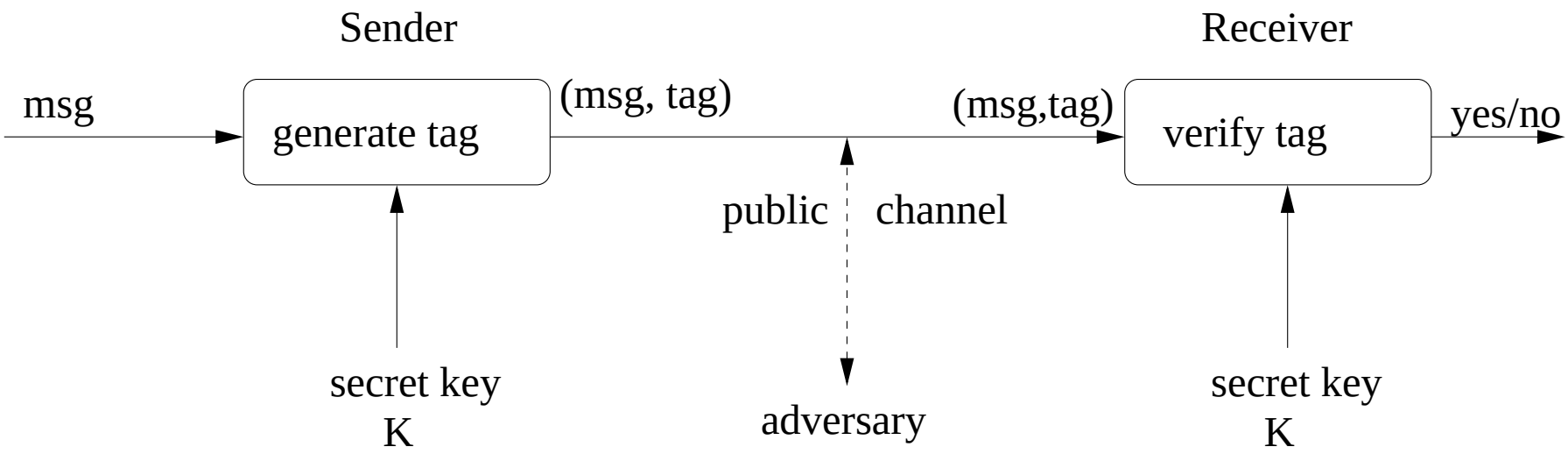
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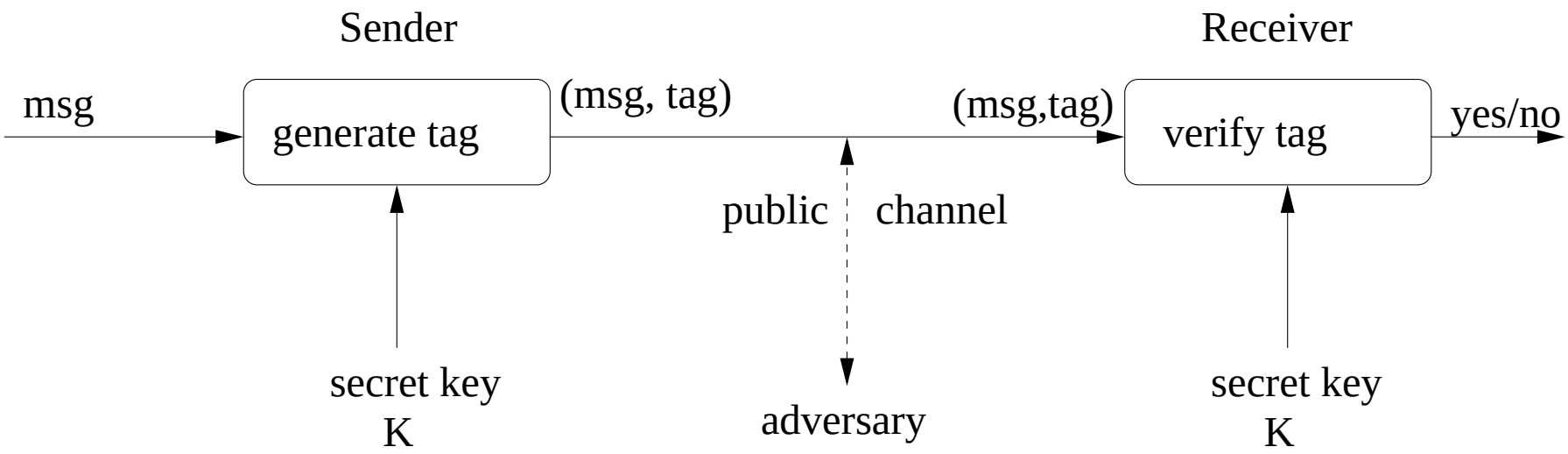
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Authenticated Encryption

- Usually both confidentiality and authentication are required simultaneously.
  - Confidentiality ensures privacy of the communication.
  - Authentication ensures integrity of the message.
- Internet packets.
  - Confidentiality and authentication of the body (data).
  - Header cannot be encrypted, since then the packet cannot be forwarded by intermediate routers.
  - Header has to be authenticated.
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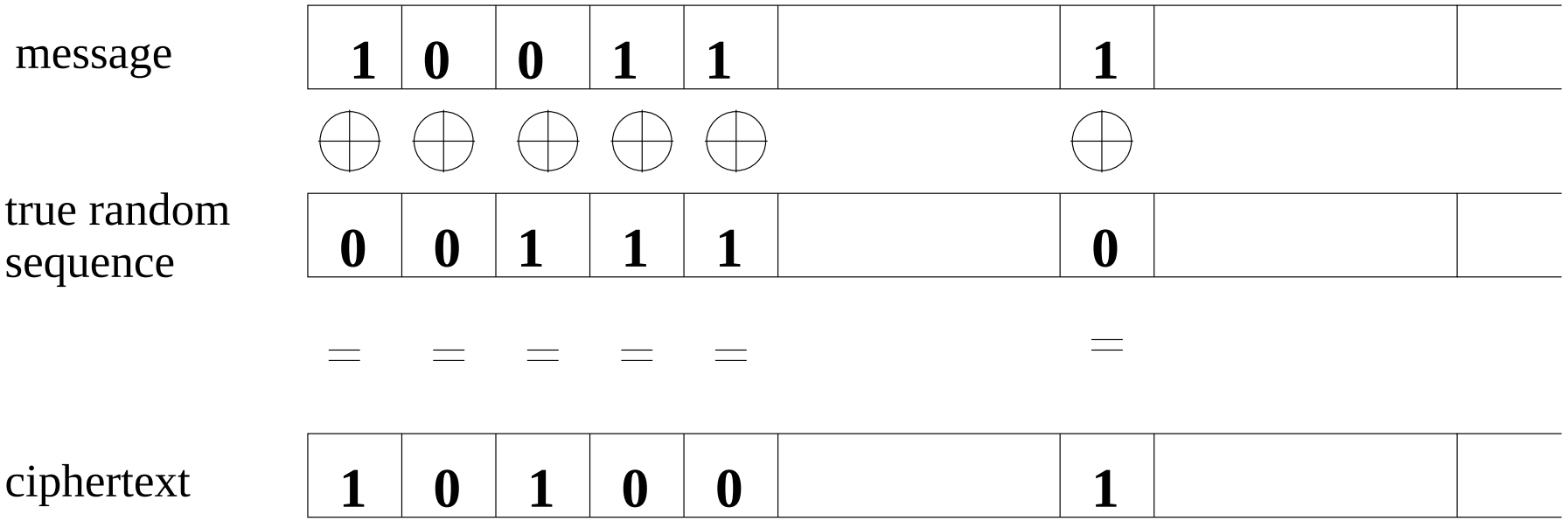


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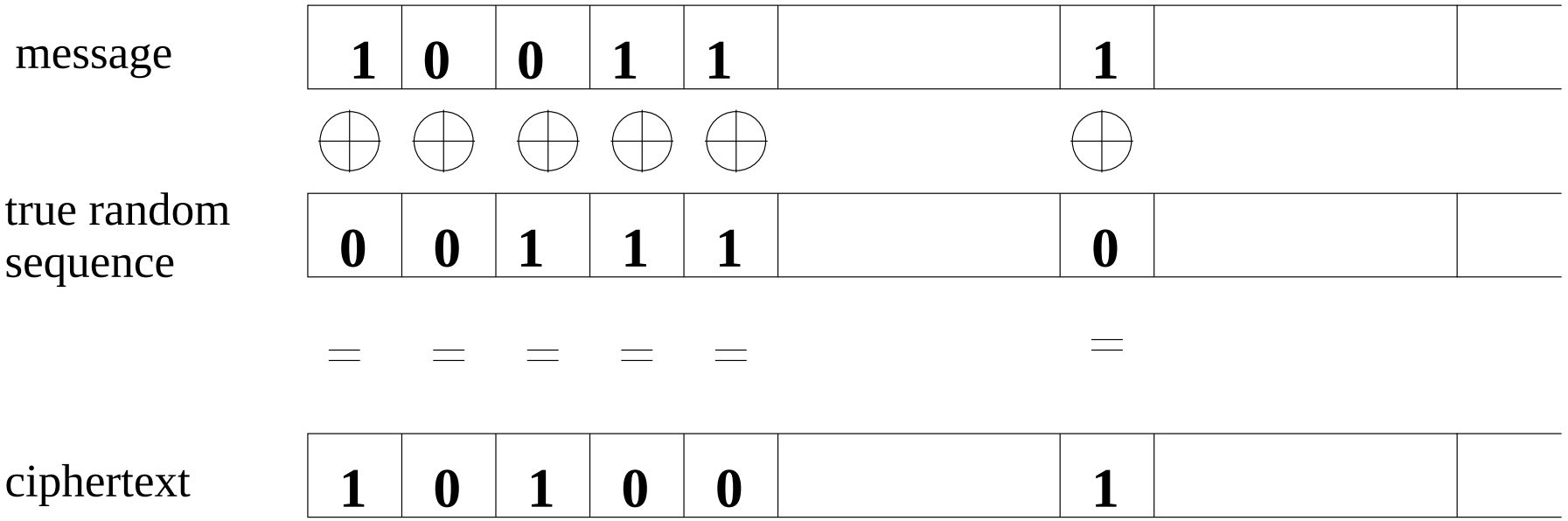


One-Time Pad



Perfect Secrecy: For $a, b \in \{0, 1\}$, $\Pr[M_i = a | C_i = b] = \Pr[M_i = a]$.

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One-Time Pad and Authentication

One-time pad does not provide authentication.

- Consider message: **attack at dawn**.
- Let $\mathbf{y}_1||\mathbf{y}_2$ be the ciphertext under one-time pad where,
  - $\mathbf{y}_1$ is the encryption of **attack**.
  - $\mathbf{y}_2$ is the encryption of **at dawn**.
- Then $\mathbf{y}_1$ itself is a valid forgery.
- If the adversary can successfully truncate the message, then the field unit may be lead into believing that the order is to attack immediately.



Basic Encryption Primitives

- Block cipher.
- Stream cipher.
- Stream cipher supporting an initialisation vector (IV).

In this talk, we focus on block ciphers.



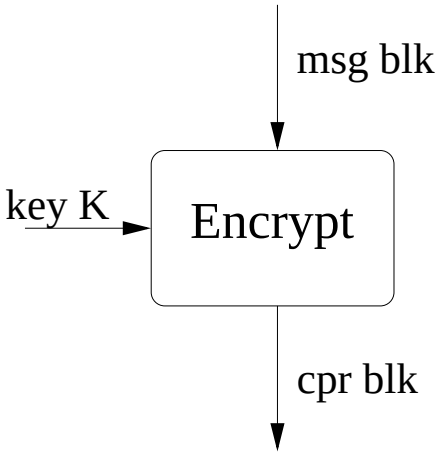
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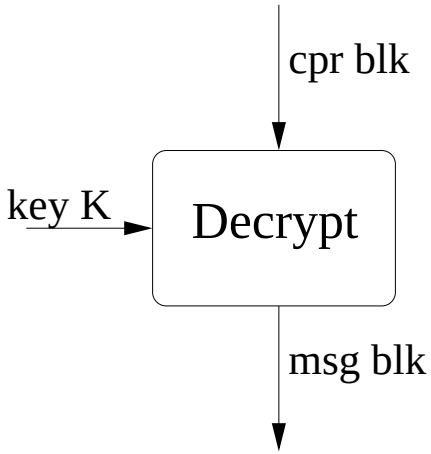
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Encrypting Short Fixed Length Strings





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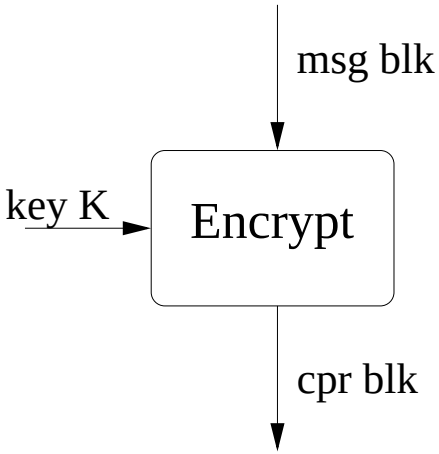
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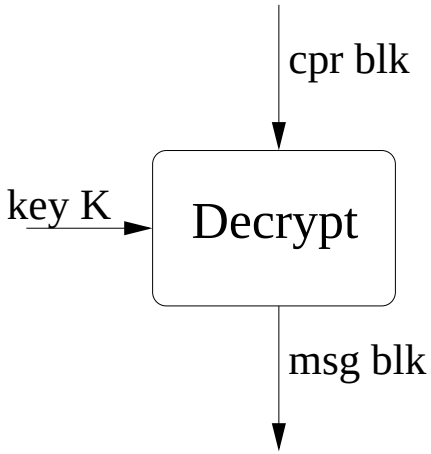
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- For each $K \in \{0,1\}^k$, $D_K(E_K(M)) = M$.
- **Typical values:** $n, k = 128, 192, 256$.



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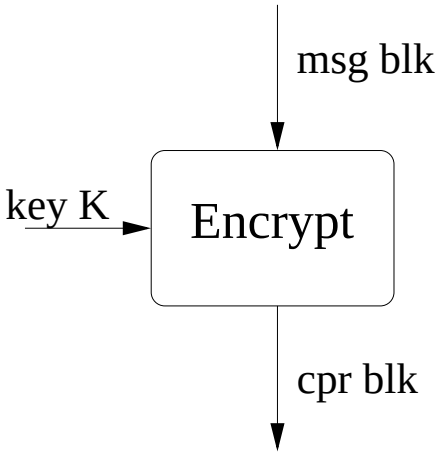
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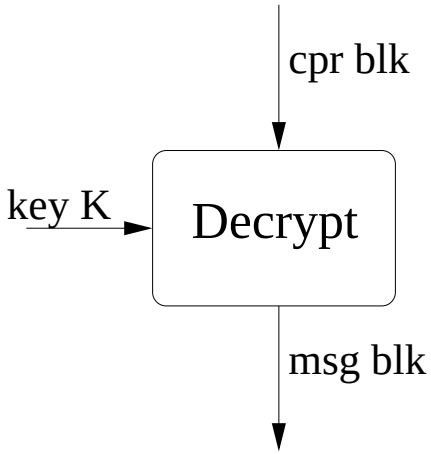
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Choosing a Block Cipher

- **Public domain approach**: AES, DES, IDEA, MARS, SERPENT, RC6, ...
  - AES: standardised by the NIST (USA) for encryption of “sensitive (unclassified)” information; FIPS 197, November 26, 2001.
- **Proprietary approach**:
  pros: benefit from “security by obscurity”;
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- **Mixed approach**:
  - start from a public domain block cipher;
  - introduce additional structure and generalize to a large family of block ciphers;
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- **Resists "known" attacks**.
  - Linear and Differential Attacks.
  - Slide, Rectangle, Impossible Differential, Boomerang and other attacks.
- **Pseudo-random permutation (PRP)**: indistinguishable from a random permutation where the adversary is allowed to ask for encryptions.
- **Strong pseudo-random permutation (SPRP)**: indistinguishable from a random permutation where the adversary is allowed to ask for encryptions *and decryptions*.



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Looking Beyond a Block Cipher

Assumption: the block cipher is “perfectly secure” (whatever that may mean).

Is that the end of the story?

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Block Cipher and Practical Requirements

Block length.

- The block cipher handles n-bit blocks.
- Typically, $n = 128, 192$ or 256 bits.

Message Requirements.

- Handle “long” messages.
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- Privacy only.
- Authentication.
- Authenticated encryption.
- Authenticated encryption with associated data.
- Others.



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Domain Extension

Electronic codebook mode (ECB): a simple approach.

- Format the message into n-bit blocks.
- Encrypt each block separately using the block cipher.

Message: $M_1, M_2, M_3, \ldots$;
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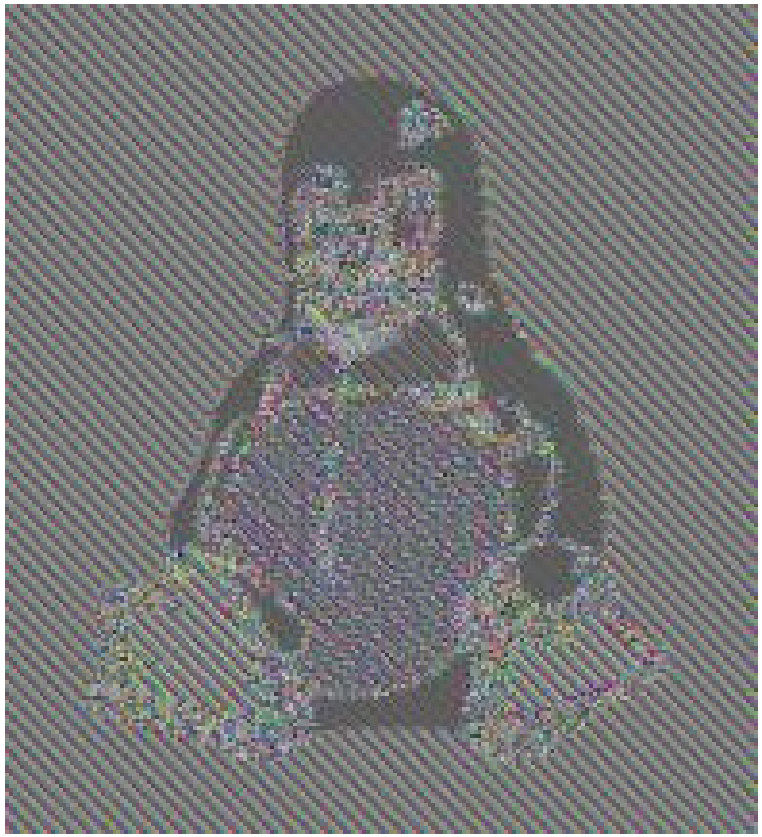
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Insecurity of ECB Mode





Source: Wikipedia

http://en.wikipedia.org/wiki/Block_cipher_modes_of_operation.

Achieving Confidentiality

Counter Mode: a simple (unrefined) description.

- Format the message into n-bit blocks.
- To encrypt the i-th block, XOR with the encryption of $R + i$.
- $R = E_K(N)$, where N is a nonce (fresh for every message).

Message: $M_1, M_2, M_3, \ldots$;
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Authentication: A Simple Scheme

Let $\{H_\tau\}_{\tau \in \mathcal{T}}$ be a family of functions such that

- for each τ, $H_\tau : \mathcal{M} \to \{0,1\}^n$;
- for a uniform random τ, and any two distinct M_1, M_2,

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- ε-almost universal hash family.

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Construction of Suitable $\{H_\tau\}_{\tau \in \mathcal{T}}$

Let $\mathcal{T} = GF(2^n)$; $\mathcal{M} = GF(2^n)^s$, i.e.,

- τ is an n-bit string;
- a message consists of s n-bit strings.

Write $M = (M_0, \ldots, M_{s-1})$. Then

$$H_\tau(M) = M_0\tau^{s-1} \oplus M_1\tau^{s-2} \oplus \cdots \oplus M_{s-2}\tau \oplus M_{s-1}.$$

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Authenticated Encryption: Constructions



Two-Pass AE Schemes

- First pass over the message to encrypt the message.
- Second pass over the ciphertext to generate a tag.
- Textbook scheme: first pass: counter; second pass: CBC; cost per block: **2[BC]**.
- Getting things right is non-trivial.
- Concrete scheme:
  - GCM (Viega-McGrew):
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Single-Pass AE Schemes

- Makes only one pass over the message.
- Cost per block: **1[BC]+**SOMETHING.
- The SOMETHING depends on the scheme.

Patented constructions.

- IACBC, IAPM (Jutla).
- XCBC, XECB (Gligor-Donescu).
- OCB (Rogaway): performs AEAD.
  - Based on the theory of tweakable block ciphers which was introduced by Liskov-Rivest-Wagner.
  - previously best known: fastest and simplest.

Constructions without patents.

- Chakraborty-Sarkar: based on general construction of TBC.
- Sarkar (2010): based on a simple and direct approach. Currently best known algorithm for AEAD.



Single-Pass AE Schemes

- Makes only one pass over the message.
- Cost per block: **1[BC]+**SOMETHING.
- The SOMETHING depends on the scheme.

Patented constructions.

- IACBC, IAPM (Jutla).
- XCBC, XECB (Gligor-Donescu).
- OCB (Rogaway): performs AEAD.
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OCB: A Brief Description

- Message is formatted into blocks $M_1, M_2, M_3, \ldots$.
- Possibly partial last block is handled properly.
- Message block M_i (if not the last one) is processed as follows.

$$C_i = \Delta_i \oplus E_K(M_i \oplus \Delta_i).$$

- $\Delta_i = x^i R(x) \bmod \tau(x)$ is the i-th mask.
  - $R = E_K(N)$, N is the nonce;
  - $\tau(x)$ is a primitive polynomial of degree n over $GF(2)$.
  - "Powering-Up" operation: can be seen as an LFSR map.



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OCB: A Brief Description (Contd.)

- Mask for the last block is obtained by multiplying the previous mask by either $(x \oplus 1)$ or by $(x^2 \oplus x \oplus 1)$ (modulo $\tau(x)$).
- The TBC theory requires the discrete logs of $(x \oplus 1)$ and $(x^2 \oplus x \oplus 1)$ to be large and well-separated.
  - Let s be such that $x^s \equiv (x \oplus 1) \bmod \tau(x)$.
  - Let t be such that $x^t \equiv (x^2 \oplus x \oplus 1) \bmod \tau(x)$.
  - Then s and t have to be large and well-separated.
- Rogaway shows one $\tau(x)$ for each value of $n = 64, 80, 128$.
- For $n = 256$, OCB (in its present form) is not defined.
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PAE: Sarkar 2010

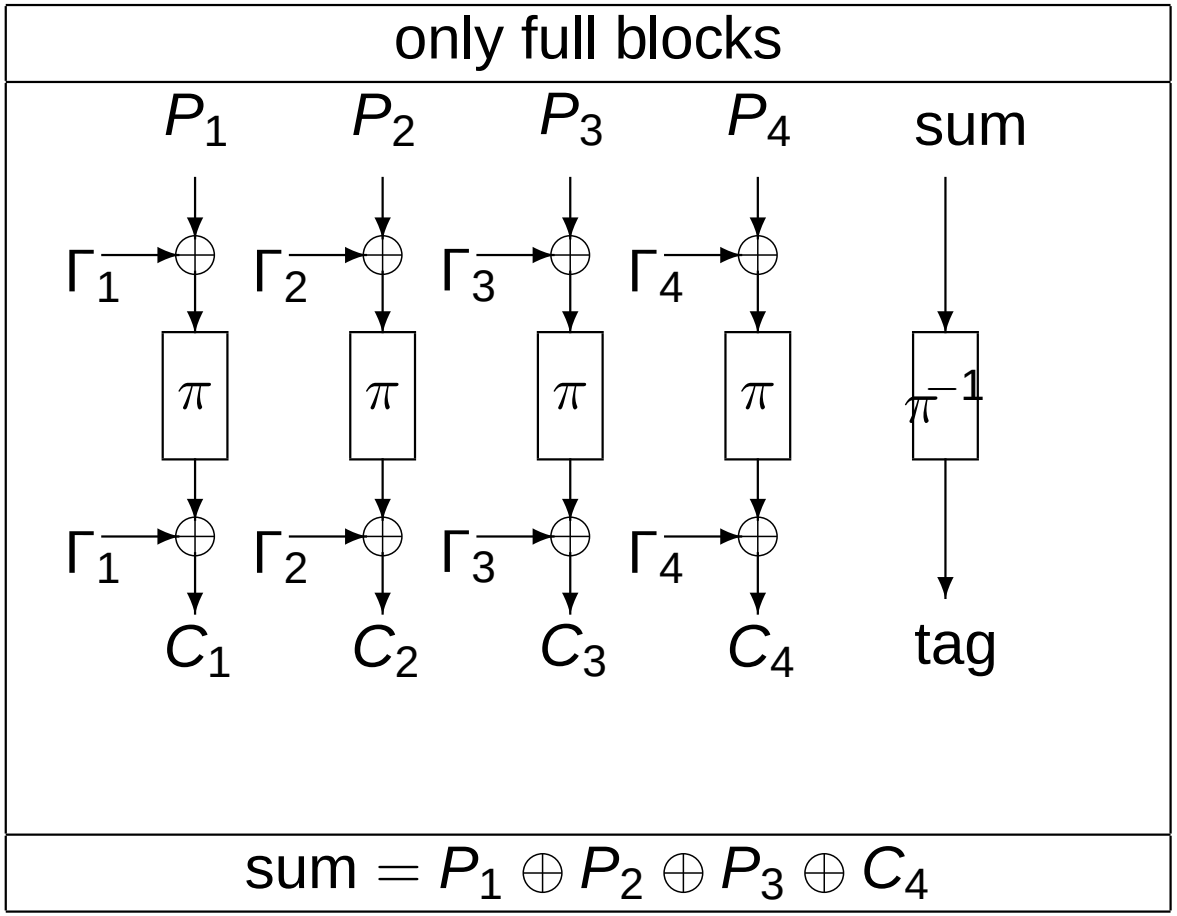


Figure: Encryption using PAE: $\gamma = \pi^{-1}(N)$; $\Gamma_i = \phi_\gamma(i)$.



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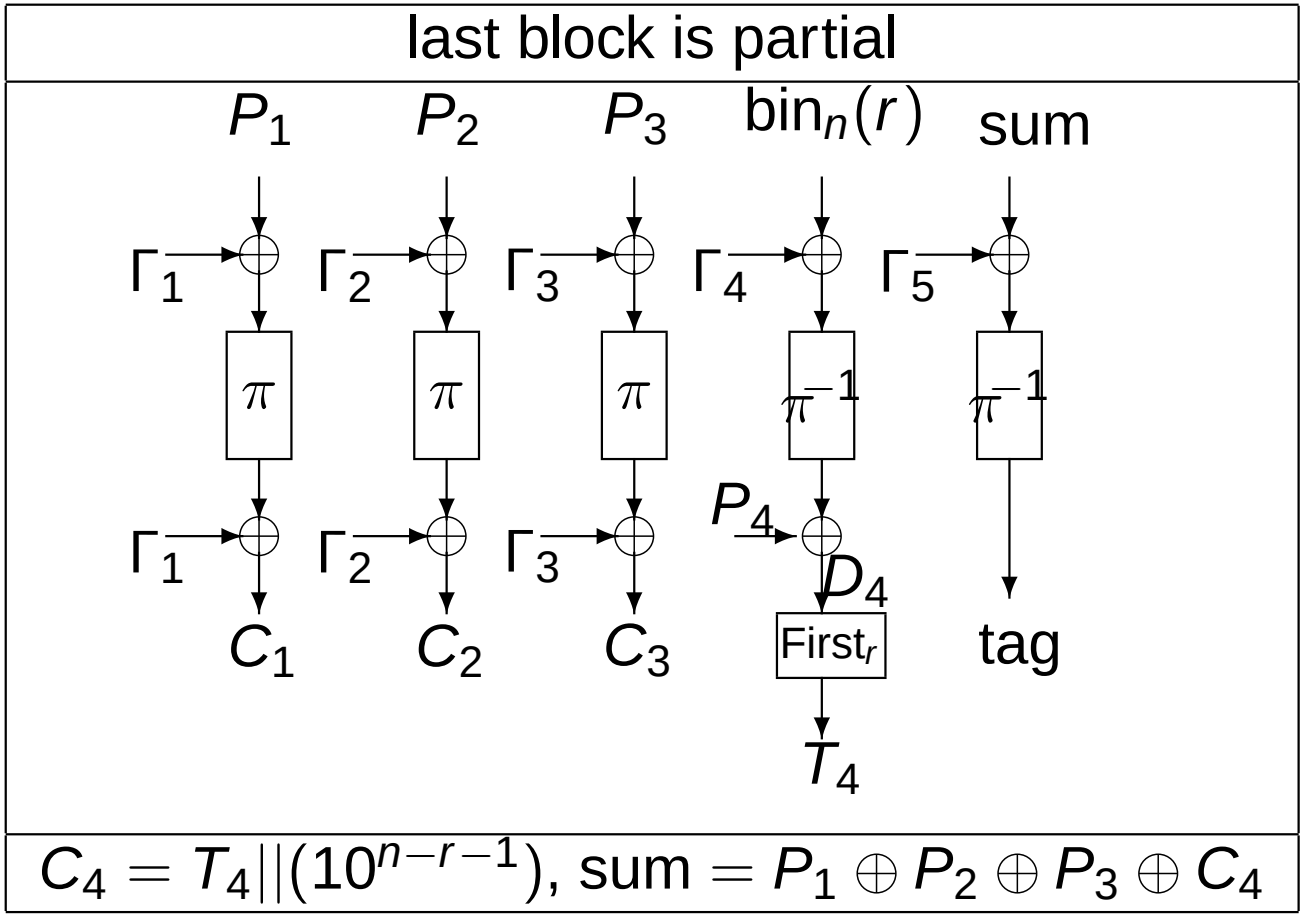


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- $\phi_\gamma(i) = \psi^i(\gamma)$ where

$$\psi : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}.$$

  - The minimal polynomial of ψ over $\mathbb{F}_2$ is a primitive polynomial.
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Advantages over OCB (and others).

- Easy to change the masking functions.
  - Simply choose another suitable ϕ.
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- Another proposal called PAE-1 is given.
  - Features similar to those of PAE.
  - The encryption algorithm of PAE-1 requires only $E_K()$.
  - Leads to smaller size hardware for encryption.
- Both PAE and PAE-1 have been extended to obtain AEAD schemes PAEAD and PAEAD-1.
- It is shown how to use PAE (or PAE-1) to obtain only an authentication scheme.
- A single module (hardware/software) can be used for different tasks.
  - Authentication.
  - Authenticated encryption.
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Conclusions

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Thank you for your attention!

